

- **Problem Statement ID :** 25013
- **Problem Statement Title:** Real-time Public Transport tracking for small cities
- **Theme:** Transportation & Logistic
- **PS Category:** Software
- **Team ID:** 52552
- **Team Name:** NextStop

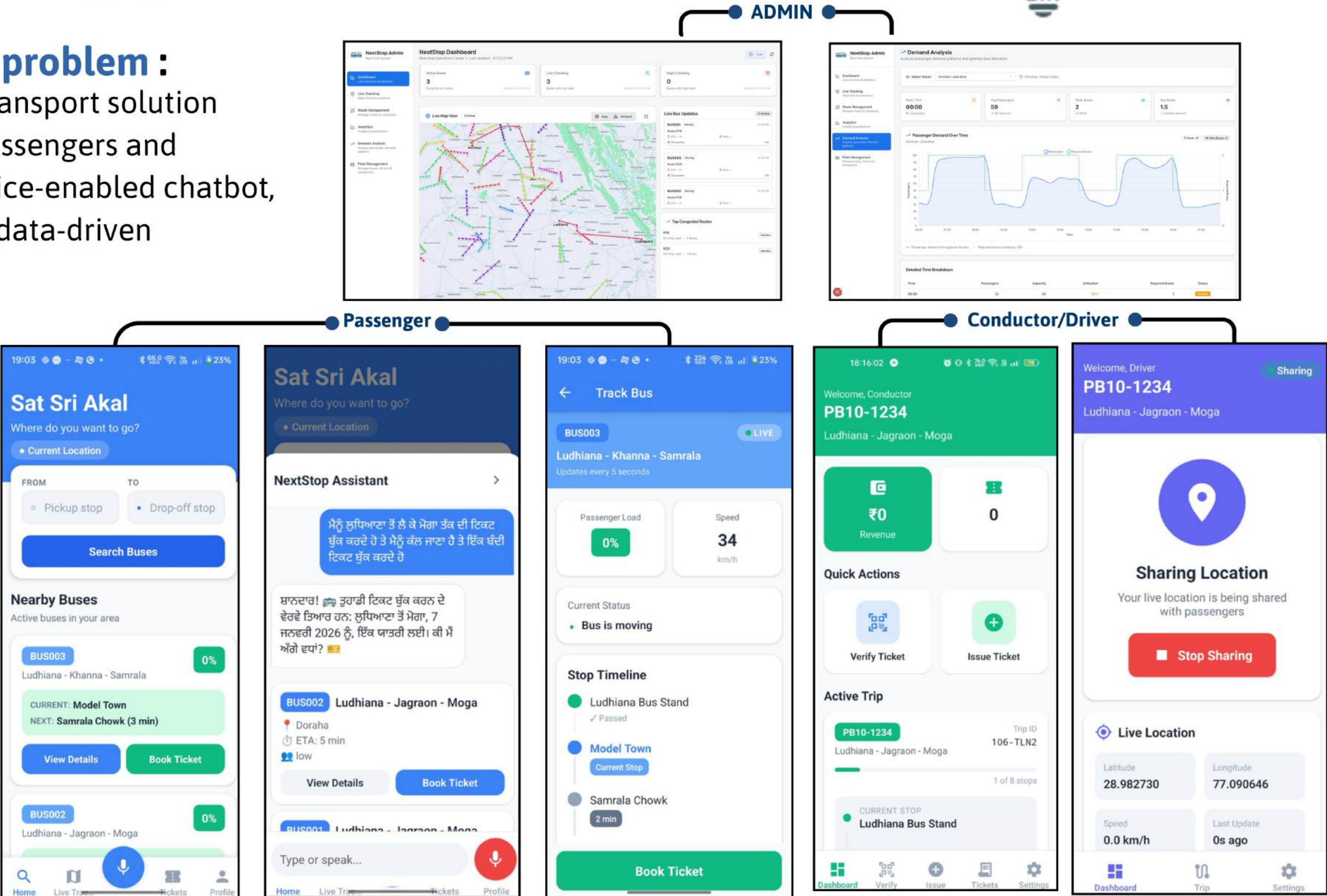


# Approach Details

**How we have solved the given problem :**  
We designed a comprehensive public transport solution comprising separate applications for passengers and conductors/drivers, a multilingual AI voice-enabled chatbot, and a centralized dashboard to enable data-driven government governance.

## Key Features:

- **Separate Conductor & User Side App**
- **Real-Time Bus Tracking**
- **Smart Route & ETA Finder**
- **Live Seat Occupancy**
- **Waiting List & Boarding Alerts**
- **Online Ticket Booking**
- **Central Monitoring Dashboard for gov.**
- **Adv. Analytics based on user data**
- **Multilingual Feature Integration**
- **Chat-bot with Voice Feature**
- **Custom suggestions from data**



# Workflow Demo



Passenger

NS

Welcome Back

Login to your account

Mobile Number

+91 Mobile Number

Get OTP

For demo, any 10-digit number works

By continuing, you agree to our Terms & Privacy Policy

1234567890

ABCDEF

GHIJKL

MNOPQRST

UVWXYZ

\*/#

0+

.

✓

Conductor

←

Bus

Driver Login

Sign in to start your shift

Email

✉

e.g., john@example.com

Password

🔒

Enter your password

Sign In

qwertyuiop

asdfghjkl

zxcvbnm

?123@☺.✓

ADMIN

NextStop Admin

Real-Time System

Dashboard

Live Tracking

Route Management

Analytics

Demand Analysis

Fleet Management

Route Management

Manage bus routes and optimize the network

Overview

Routes Data

Schedules

Optimization

Amritsar-Jalandhar

Amritsar Bus Stand → Jalandhar Bus Stand

25 - km

Stops: Amritsar Bus Stand, Verka Chowk, Majitha Chowk, Batala, Qadian, Kala Bakra, Jalandhar Bus Stand

Daily Passengers: 0

Peak Occupancy: 50%

Last Updated: 1/6/2026

optimal

Moga-Mansa

Moga Bus Stand → Mansa Bus Stand

25 - km

Stops: Moga Bus Stand, Mansa Crossing, Patankot Stop, Sangrur Chowk, Mansa Bypass, Mukhtar Crossing, Mansa Bus Stand

Daily Passengers: 0

Peak Occupancy: 50%

Last Updated: 1/6/2026

optimal

Ropar-Mohali

Ropar Bus Stand → Mohali Bus Stand

25 - km

Stops: Ropar Bus Stand, Mohali Bypass, Mohali Terminal, Mohali Terminal, Mohali Chowk, Mohali Crossing, Mohali Stop, Mohali Bus Stand

Daily Passengers: 0

Peak Occupancy: 50%

Last Updated: 1/6/2026

optimal

Sangrur-Mohali

Sangrur Bus Stand → Mohali Bus Stand

25 - km

Stops: Sangrur Bus Stand, Jalandhar Terminal, Fatehgarh Sahib Terminal, Amritsar Circle, Kapurthala Bypass, Mohali Chowk, Sangrur Circle, Mohali Bus Stand

Daily Passengers: 0

Peak Occupancy: 50%

Last Updated: 1/6/2026

optimal

Route Statistics

Total Routes: 70

Total Daily Passengers: 0

Average Peak Occupancy: 50%

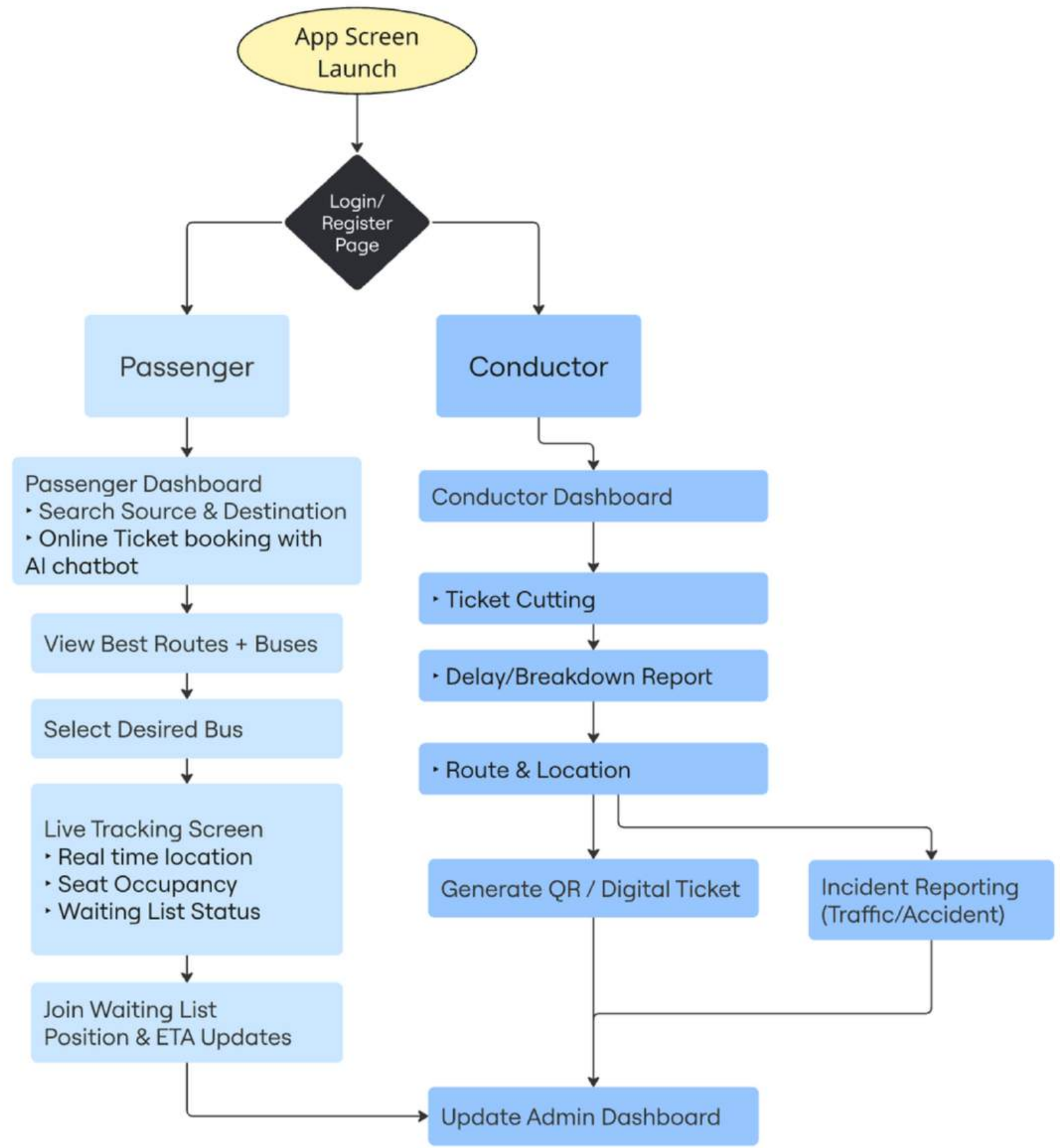
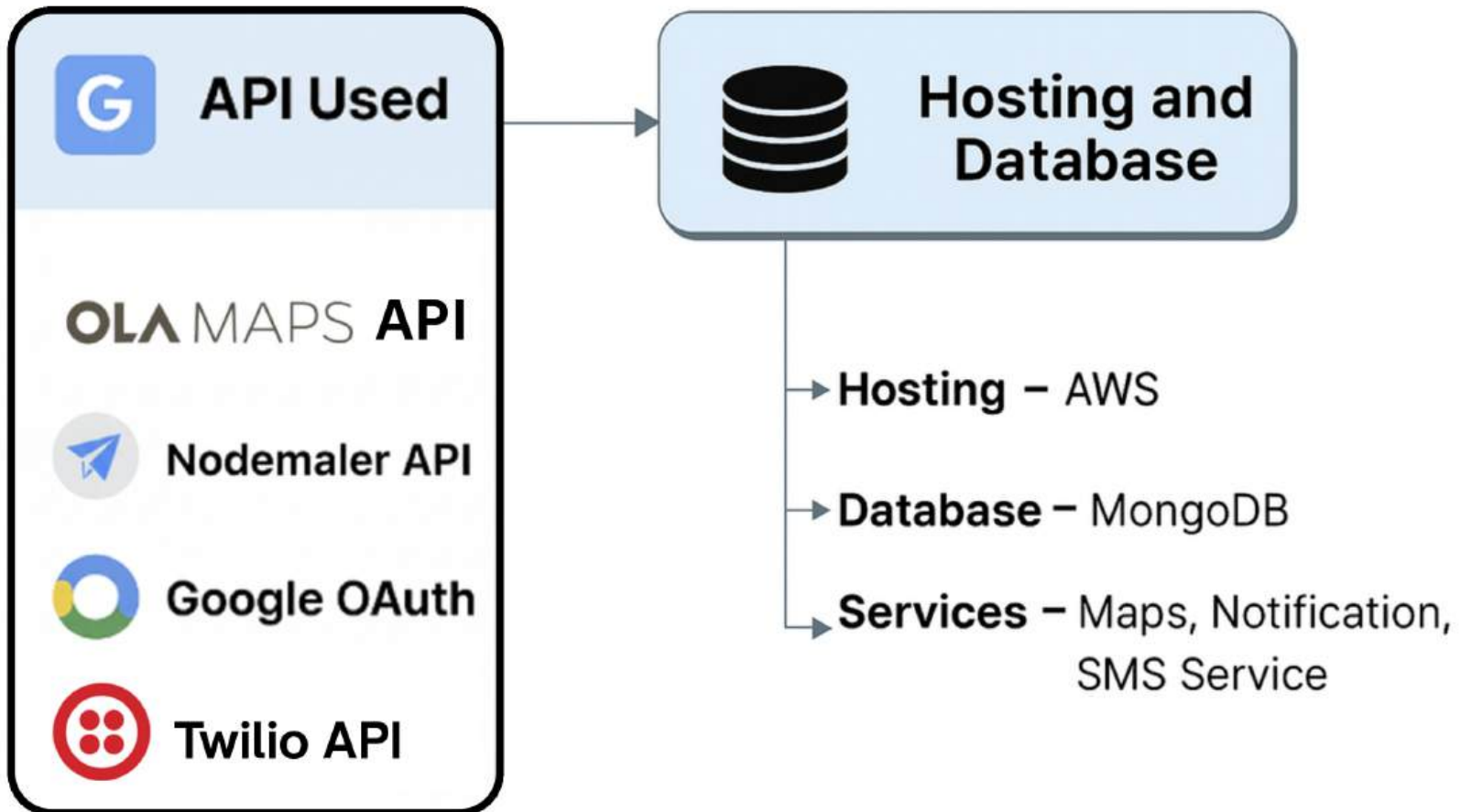
0:10

🔊

🗑️

# Technical Approach

## Tech Stack Used



# Feasibility & Viability



## Feasibilities



### Real Time GPS Bus Tracking

Improves ETA accuracy by up to 80–85%



### Smarter Route Planning

Reduces Congestion by upto 15–20%



### Live Passenger Count

Prevents Overcrowding



### AI Voice Agent (MCP)

Listens to user voice input and automatically fills journey details in real time.

## Challenges



### Connectivity Gaps in Rural Areas

40% of rural depots face poor internet coverage



### Legacy ETM Machines

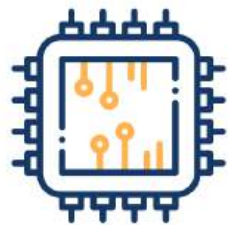
ETMs are outdated, lacking real-time sync



### Language barrier for people

Punjab have many people who do not understand punjabi and many who do not understand english

## Strategies



### Multi-Lingual

Hindi,English and **Punjabi** language available for user



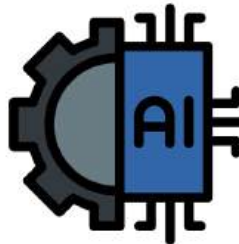
### Unified Digital Platform

single dashboard for all state + private bus operators



### Email services

Email-based updates ensure inclusivity



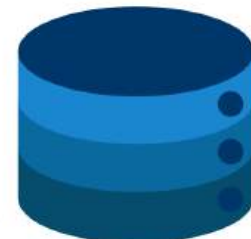
### AI-driven analytics

Predicts delays, optimizes routes

# Impact & Benefits



**Reduced Waiting Times**  
Commuters can easily know the exact arrival times of buses in advance



**Better Data for Authorities**  
Real-Time demand helps optimise routes and frequency



**Digital Ticket Booking with AI Support**  
Allows commuters to plan and book tickets quickly using an AI chatbot.



**Multi-Lingual in Punjabi and Hindi**  
Voice-enabled ticket booking in Punjabi, with full multilingual platform support.



**Less Overcrowding**  
Passenger Count info helps commuters choose comfortably



**Improved Safety**  
Real-time bus location sharing enhances travel security and peace of mind.



# Research & References



- [Malfunctioning Ticket Machines a Big Hassle, Say Bus Conductors](#)  
– Hindustan Times – Kriti, Rishika (2025)
- [Smart Bus Seat Occupancy Detection using IoT](#)  
– International Research Journal of Modernization in Engineering Technology and Science – Vishwasrao, S., Sapkal, A., Yeole, K., and Sonawane, S.S. (2021)
- [Bus Karo – Live Bus Tracking System](#)  
– IJRASET (Journal for Research in Applied Science & Engineering Technology) – Bansal, E., Gupta, P., Sharma, U., Dandona, V., and Jain, D. (2024)
- [Real-Time Tracking System for City Bus Location and Passenger Count](#)  
– International Journal of Advanced Research in Science, Communication and Technology (IJARSCT) – Gade, S.; Pandit, R.B.; Pawara, K.; Pinjari, T.; Patil, G.; and Datir, P. (2024)
- [To Each Route its Own ETA: A Generative Modeling Framework for ETA Prediction](#)  
– arXiv Preprint – Charul, and Biyani, P. (2019)
- [AWS Cloud Case Studies for Transportation](#)